Supplementary Material: Appendices for "Thermodynamic approach and Saddle-Dominated Capacity of the LSE Dense Associative Memory"

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A Appendix A: Derivation of the prefactor in the non-retrieval sum

The Debye uniform asymptotic of the modified Bessel function (Olver et al. 2010, 10.41.3) gives the prefactor of Eq. (13). Keeping the leading ($k = 0$) term of the asymptotic series,

$$I_\nu(\nu z') \sim \frac{\mathrm{e}^{\nu\eta(z')}}{(2\pi\nu)^{1/2}(1+z'^2)^{1/4}}, \tag{36}$$

where $z' = 2\beta N/(N-2)$ and

$$\eta(z') = \sqrt{1+z'^2} + \ln\frac{z'}{1+\sqrt{1+z'^2}}\,. \tag{37}$$

$\eta(z')$ is expanded to $\mathcal{O}(N^{-1})$ because it is multiplied by $\nu = N/2-1$ in the exponent:

$$\eta(z') = \eta(2\beta) + \eta'(2\beta)\frac{4\beta}{N} + O(N^{-2})\,, \tag{38}$$

where

$$\eta(2\beta) = 1 + 2\beta\varphi^* + \ln\varphi^*\,, \tag{39}$$

$$\eta'(2\beta) = \varphi^* + \frac{1}{2\beta}\,. \tag{40}$$

Then

$$\nu\eta(z') = \tfrac{N}{2}\eta(2\beta) - \ln\varphi^* + \mathcal{O}(1/N)\,. \tag{41}$$

The $\mathcal{O}(1)$ piece $-\ln\varphi^*$ in the exponent contributes a factor $1/\varphi^*$ to the prefactor. Using Stirling's formula

$$\Gamma(N/2) \sim \sqrt{4\pi/N}\,(N/2)^{N/2}\,\mathrm{e}^{-N/2} \tag{42}$$

and collecting all terms in Eq. (18) to $\mathcal{O}(1)$ in the exponent and to leading order in the prefactor reproduces the exponent of Eq. (13) and gives the prefactor:

$$S_{\mathrm{nr}} \approx \frac{\mathrm{e}^{N[\alpha+g_{\max}(\beta)-\beta]}}{\sqrt{1-\varphi^{*4}}}\,, \tag{43}$$

with $g_{\max}(\beta)$ given by Eq. (12). At $\beta = 1$, $\varphi^* \approx 0.618$ and the prefactor is ≈ 1.082, reproducing Eq. (14).

B Appendix B: Tangency of the cusp lines to the spherical extreme

To show that the cusp lines (16) are tangent to the spherical-extreme curve (6) at every β, we need to show that the slope and the φ-intercept coincide.

Differentiating (6) once gives

$$\varphi'_{\max}(\alpha) \;=\; \frac{1-\varphi_{\max}^2}{\varphi_{\max}}\,, \tag{44}$$

and a second differentiation gives $\varphi''_{\max}(\alpha) < 0$ for $\alpha > 0$, so $\varphi_{\max}$ is strictly concave on $(0,\infty)$. For a concave function every tangent line lies above the function, with equality only at the tangency point. This is used in the lower-envelope identity below.

The cusp line at sharpness β,

$$\varphi \;=\; \alpha/\beta + g_{\max}(\beta)/\beta\,, \tag{45}$$

has slope $1/\beta$ and φ-intercept $g_{\max}(\beta)/\beta$. To find the point on the blue curve where the tangent has the same slope, set the right-hand side of (44) equal to $1/\beta$:

$$\frac{1-\varphi^2}{\varphi} \;=\; \frac{1}{\beta} \quad\Longleftrightarrow\quad \beta\,\varphi^2 + \varphi - \beta = 0\,. \tag{46}$$

This is the saddle equation and its unique positive root is the spherical saddle (11). The corresponding α on the blue curve follows from $\varphi^* = \varphi_{\max}(\alpha^*)$:

$$\alpha^*(\beta) \;=\; -\tfrac{1}{2}\ln(1-\varphi^{*\,2})\,. \tag{47}$$

A line of slope $1/\beta$ passing through $(\alpha^*(\beta), \varphi^*(\beta))$ has φ-intercept

$$b_{\mathrm{tg}}(\beta) \;=\; \varphi^* - \alpha^*/\beta\,. \tag{48}$$

The cusp line has φ-intercept $g_{\max}(\beta)/\beta$. Using (12), the φ-intercepts coincide; the two lines are therefore identical.

By the concavity statement above, the cusp line at any sharpness β lies above the blue curve, with equality only at $\alpha = \alpha^*(\beta)$:

$$\varphi_c(\alpha,\beta) \;\geq\; \varphi_{\max}(\alpha) \quad \text{for all } \alpha \text{ and all } \beta > 0\,, \tag{49}$$

with equality at one α for each β. Taking the infimum over β,

$$\varphi_{\max}(\alpha) \;=\; \inf_{\beta>0}\varphi_c(\alpha,\beta)\,, \tag{50}$$

the lower-envelope identity.

The saddle function $\varphi^*(\beta)$ (11) is continuous and strictly increasing on $(0, \infty)$ from 0 to 1. As β sweeps $(0, \infty)$, the tangency point $(\alpha^*(\beta), \varphi^*(\beta))$ therefore traces the entire blue curve from the origin to its right edge.

Two results follow from the geometric setup above. At $T = 0$ the cusp line of slope $1/\beta$ reaches $\varphi = 1$ at $\alpha = \beta - g_{\max}(\beta)$, which is the zero-temperature capacity (17) (equivalently, the spinodal at $T = 0$).

The Ramsauer et al.'s condition $\varphi_{\max}(\alpha) = 1 - \alpha/\beta$ is the intersection of $\varphi_{\max}$ with the line of slope $-1/\beta$ through $(0, 1)$. By (50), $\varphi_{\max}$ lies below every cusp line, so the Ramsauer et al.'s line crosses $\varphi_{\max}$ at strictly smaller α than the cusp line at slope $1/\beta$ reaches $\varphi = 1$. Ramsauer et al.'s capacity α_c^R therefore satisfies $\alpha_c^R < \alpha_c(\beta, 0)$ for every $\beta > 0$.

Using (11) in (12) and then in (17), one obtains:

$$
\begin{aligned}
\alpha_c(\beta, 0) &= \beta - g_{\max}(\beta) \;\to\; \beta & (\beta \to 0)\,, \\
\alpha_c(\beta, 0) &= \beta - g_{\max}(\beta) \;\to\; \tfrac{1}{2}\ln\beta + \tfrac{1}{2} & (\beta \to \infty)\,,
\end{aligned}
\tag{51}
$$

the small- and large-β limits of the zero-temperature capacity cited in the Abstract.

Legendre transform. The lower-envelope identity (50) and the formula below are the two sides of a Legendre transform between $\varphi_{\max}(\alpha)$ and $g_{\max}(\beta)/\beta$. The forward transform takes $\varphi_{\max}$ to its conjugate,

$$
g_{\max}(\beta)/\beta \;=\; \sup_{\alpha} \big[\, \varphi_{\max}(\alpha) - \alpha/\beta \,\big]\,, \tag{52}
$$

with the supremum attained at $\alpha = \alpha^*(\beta)$ from the slope-match condition (46). The Legendre transform is involutive: applying it to $g_{\max}(\beta)/\beta$ recovers $\varphi_{\max}(\alpha)$, which is exactly (50) after using the cusp-line definition (16).

Two facts follow from the duality. First, the lower-envelope identity (50) is the inverse of (52), so the tangency is automatic. Second, differentiating (12) in β and using the envelope theorem $\partial g/\partial\varphi|_{\varphi^*} = 0$ gives

$$
g'_{\max}(\beta) \;=\; \varphi^*(\beta)\,, \tag{53}
$$

so $g_{\max}(\beta)$ is recovered by integrating $\varphi^*(\beta)$. Plugging the small- and large-β limits $\varphi^* \sim \beta$ and $\varphi^* \sim 1 - 1/(2\beta)$ into (53) and integrating reproduces (51).

References

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